9-30

ch 5 § 7

Recall wave eq'n

$u_{xx} - u_{yy} = 0$ ← comes from physics

$\xi = x+y \qquad \eta = x-y$ ~~[illegible]~~ (a rotation + reflection)

$\Rightarrow u_{\xi\eta} = 0$ ← easy to solve!

solve $u_{\xi\eta} = 0$

$$\int u_{\xi\eta}\, d\eta = C_1(\xi)$$

$$u_\xi = C_1(\xi)$$

$$\int u_\xi\, d\xi = \int C_1(\xi)\, d\xi$$

$$u = F(\xi) + C_2(\eta)$$

$$u(\xi,\eta) = F(\xi) + G(\eta)$$

$$\boxed{u(x,y) = F(x+y) + G(x-y)}$$ d'Alembert sol'n

check

$u_x = F'(x+y)(1) + G'(x-y)(1)$

$u_{xx} = F''(x+y)(1) + G''(x-y)(1)$

$u_y = F'(x+y)(1) + G'(x-y)(-1)$

$u_{yy} = F''(x+y) - G''(x-y)(-1)$

$u_{xx} - u_{yy} = 0$ ✓

$F(x+y) = \sin(x+y)$

Ch 6 Eq'ns of Mathematical Physics

§6.1 Divergence Thm & Green's Identities

vector field $\underline{V} = (P, Q, R) \in \underbrace{[C^1(\Omega)]^3}_{P, Q, R \in C^1(\Omega)}$

$$\operatorname{div}(\underline{V}) = \underline{\nabla} \cdot \underline{V} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \langle P, Q, R \rangle$$

$$= \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

We assume Ω is open, connected, bounded, and has Lipschitz boundary (think finite # of corners). We call this a <u>regular domain</u>.

Divergence Thm

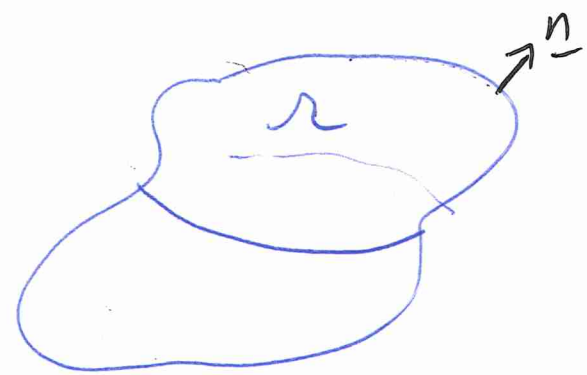
If $P, Q, R \in C^1(\Omega) \cap C^0(\bar{\Omega})$ and Ω is regular, (closure)

Then

$$\underbrace{\int_\Omega \underline{\nabla} \cdot \underline{V} \, d\underline{x}}_{\text{volume integral}} = \underbrace{\int_{\partial\Omega} \underline{V} \cdot \underline{n} \, d\sigma}_{\text{surface integral}}$$

where $\underline{n}$ is the unit normal pointing out.

Ref: Thm 1.5.3.1 of Grisvard
"Elliptic Problems in nonsmooth Domains"



What is the Divergence Thm in 1D?

$\Omega = (a,b) \qquad \underline{v} = p(x) \qquad \partial\Omega = \{a,b\}$

$$\int_a^b p'(x)\,dx = \sum_{i=a,b} p \cdot \underline{n}$$

$$= p(x=a)(-1) + p(x=b)(+1)$$

$$= p(b) - p(a)$$ Fundamental Thm of calc!

<u>Notation</u>

$$\Delta u = \underline{\nabla}^2 u = \underline{\nabla} \cdot \underline{\nabla} u$$

$$= \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle u$$

$$= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) u = u_{xx} + u_{yy} + u_{zz}$$

$\Delta = \underline{\nabla}^2 = \underline{\nabla} \cdot \underline{\nabla} =$ <u>Laplace operator</u>

$\Delta u = 0$ the <u>Laplace eq'n</u>

Identity

$$\underbrace{u\,\underbrace{\underline{\nabla}^2 w}_{\text{scalar}}}_{\text{scalar}} = \underbrace{\underline{\nabla}\cdot\underbrace{(u\,\underbrace{\underline{\nabla} w}_{\text{vector}})}_{\text{vector}}}_{\text{scalar}} - \underbrace{\underbrace{\underline{\nabla} u}_{\text{vector}}\cdot\underbrace{\underline{\nabla} w}_{\text{vector}}}_{\text{scalar}}$$

sanity check

Pf HW $\quad \underline{\nabla} = \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \rangle$

$w(x,y,z) \quad u(x,y,z)$

$\Delta w = u_{xx} + u_{yy} + u_{zz}$

Integrate identity

$$\int_\Omega u \underline{\nabla}^2 w \, d\underline{x} = \underbrace{\int_\Omega \underline{\nabla} \cdot (u \underline{\nabla} w) d\underline{x}}_{= \int_{\partial\Omega} u \underline{\nabla} w \cdot \underline{n} \, d\sigma} - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w \, d\underline{x}$$

$\frac{\partial w}{\partial \underline{n}} = \underline{\nabla} w \cdot \underline{n}$ how does w change in the direction of $\underline{n}$

$$\boxed{\begin{array}{l} \underline{\text{Green's First Identity}} \\ \int_\Omega u \underline{\nabla}^2 w \, dx = \int_{\partial\Omega} u \frac{\partial w}{\partial \underline{n}} d\sigma - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w \, d\underline{x} \end{array}}$$

$$\int u \, dv = uv - \int v \, du$$

$v = \underline{\nabla} w$

<u>Assumption</u> Ω is regular, $u, w \in C^1(\Omega) \cap C^0(\bar{\Omega})$

$\int_\Omega u \underline{\nabla}^2 w \, d\underline{x}$ must be finite